Effects of Large-Scale Orography on January Atmospheric Circulation: A Numerical Experiment

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Effects of large-scale orography on January atmospheric circulations are studied with the use of a general circulation model (the MRI •GCM). The model is a 5-layer tropospheric model with its top at 100mb. The horizontal resolution of the model is 5**4* in the longitudinal and the latitudinal directions, respectively. Four cases of experiment are performed under the perpetual January condition to isolate the orographic effects of the Tibetan Plateau and those of the Rockies and Greenland. One is the run where all mountains are retained (M). Others are the run where only the Asian mountains are removed (NAS), the run where mountains in North America and Greenland are removed (NRG), and the run where all mountains are removed (NM). Our discussions are mostly concerned with the orographic effects on the stationary fields.

Our experiment shows that large-scale mountains tend to divert flows around orographies. The ascending (descending) center shifts poleward (equatorward) in the upwind (downwind) side of mountains. As for the zonal wavenumber 1, the surface vertical motions are very different from those used as the conventional linear kinematical surface boundary condition.

Most of stationary waves in the middle latitudes are basically represented by a linear superposition of the Rossby wave-trains forced by the Tibetan Plateau and by the Rockies plus Greenland mountains. The Tibetan Plateau strengthens the anticyclonic circulation southeast of Japan and the cyclonic one over the Aleutian Islands as well as a ridge and a trough at the northwest and the southeast sides of the Plateau, respectively. The Tibetan Plateau enhances the cold surge along the eastern periphery of China down to the equatorial Indian Ocean. The Rockies plus Greenland mountains strengthen a ridge and a trough at the west and the east sides of the Rockies, respectively. The zonal scale of this response is much larger than the geographical scale of the Rockies.

In the Northern Hemisphere, wavenumber 1 stationary wave is mainly forced thermally, while those of higher wavenumbers orographically. In the Southern Hemisphere, on the other hand, thermally forced stationary waves of wavenumbers 2 to 4 are dominant in the subtropical latitudes. The Antarctica forces wavenumber 1 which has a structure similar to the thermally forced wavenumber 1 in the high latitudes of the Northern Hemisphere. Enhanced cooling over the elevated Antarctica is most responsible for this wave. This cooling difference is attributed to the differences between M and NM in the vertical distribution of clouds and/or in the parameterized snow albedo.

Effects of the Tibetan Plateau are not small in the equatorial latitudes of the Southern Hemisphere. In the extratropics of the Southern Hemisphere, however, the Rockies plus Greenland forcing causes larger responses than the Tibetan plateau forcing does. An intermittently formed westerly duct over the equatorial Atlantic is responsible for the cross-equatorial propagation of the Rockies plus Greenland effect.

In the Northern Hemisphere, orographies suppress the intensity of eastward moving waves which are effective in the poleward heat transport. On the other hand, they enhance the intensity of westward moving waves, which are almost irrelevant to the poleward heat transport.

In total, orographies suppress the poleward heat transport by transient eddies, although the transient eddy kinetic energy changes little with or without orographies. The decrease of the poleward heat transport by

Introduction

It is well known that orographies have large impacts on the atmospheric circulations. As reviewed by Kasahara (1980), large-scale orographies influence atmospheric motions of all scales from planetary waves through lee cyclones to small turbulent eddies. Charney and Eliassen (1949) demonstrated the large dynamical impacts for the stationary flow of the Northern Hemisphere winter. Many theoretical studies have been presented since that time on orographic effects in the atmospheric flows. Restricting our interest to the impacts on largescale stationary flows, we have to mention the studies of the vertical propagation of stationary planetary waves initiated by Charney and Drazin (1960). Matsuno (1970) showed, with the use of a steady linearized geostrophic potential vorticity equation in spherical co-ordinates and of the observed mean geopotential height at 500 mb as a lower boundary condition, good agreement with the observed state for the wavenumber 1 in the Northern Hemisphere winter stratosphere. Although Matsuno (1970) did not clarify the nature of forcing in the troposphere, Kasahara et al. (1973) and Manabe and Terpstra (1974) showed the importance of orography in maintaining the stationary Aleutian High in the stratosphere after the general circulation experiments with and without orography.

The studies on the horizontal propagation of Rossby waves were intensively made since the mid 1970s. Hoskins et al. (1977) studied energy dispersion on the sphere governed by a linear barotropic vorticity equation. They showed that planetary-scale vorticity sources tend to produce an orderly downstream train of waves with a horizontal tilt to the latitude circle. This implies that orographic effects should be studied not in a zonal channel flow but in a spherical domain.

Hoskins and Karoly (1981) extended the study to the response of a spherical baroclinic atmosphere to thermal and orographic forcings. In the upper troposphere, they showed that the vorticity sources generate wave-trains which are very similar to those given by a barotropic model. They also showed that such wave-trains are interpreted as stationary Rossby waves propagating in a slowly varying medium

One of the main motivations of the present study is to reconsider the orographic effects on the stationary planetary waves from the standpoint of Rossby wave propagation on a sphere with use of a general circulation model. Recently, Held (1983) presented a similar study with a GFDL spectral GCM of 15 wave rhomboidal truncation. In the present study, we would like to understand, further in detail, the effects of the Tibetan Plateau and those of the Rockies, separately. For this end, we performed four cases of experiment. One is the control run (M) where all mountains are retained. Others are the run (NAS) where mountains over the Asian continent removed, the run (NRG) where mountains over North America and Greenland are removed, and the run (NM) where the entire mountains are removed. By comparing between them, we will isolate effects of the Tibetan Plateau, and those of the Rockies plus Greenland. We also discuss the effects of the Antarctica in January.

An outline of the model and the experiment is described in section 2. In section 3, orographic effects on stationary planetary waves are presented. The differences in the zonal mean field are described in section 4. Most of the present discussions are on the stationary planetary waves. Characteristic changes in the transient waves are touched only briefly in section 5. More detailed study on them will be reported in a separate paper.

An outline of the model and the experiment

Model

The model used in the present experiment was developed at the Meteorological Research Institute (MRI) and is described by Tokioka et al. (1984). The model (MRI • GCM-I) is basically identical to the UCLA (University of California at Los Angeles) model described by Arakawa and Mintz (1974) and Arakawa and Lamb (1977), with minor changes to both the dynamical and physical processes of the model. Ground temperature, ground wetness, snow depth, atmospheric boundary layer depth and the gaps in physical variables at the top of the planetary boundary layer (PBL) are predicted by the model besides two components of horizontal wind velocity, temperature, the mixing ratio of water vapor and surface pressure. The surface albedo is determined by the model as a simple function of the surface boundary. The cloud amount, also, is determined diagnostically by the model. The January and July performances of the model are reported by Tokioka et al. (1985) and Kitoh and Tokioka (1986), respectively

Governing equations are approximated by a finite differencing method in both the horizontal and the vertical directions. A modified σ-coordinate * , $\sigma = (p - pl)/(ps - pt)$, is used in the vertical direction, where p is the pressure, pt the pressure at the top of the model (=100mb), and Ps the surface pressure. The model atmosphere is divided into five layers in the vertical direction with interfaces located at $\frac{1}{9}, \frac{3}{9}, \frac{5}{9}$ and $\frac{7}{9}$ in σ. The Arakawa C grid scheme is used with a horizontal resolution of 5○ by 6○ in the longitudinal and the latitudinal directions, respectively. The time differencing used in the model for the basic dynamical terms is the leapfrog scheme with a periodical insertion of the Matsuno scheme.

In the computation of radiative heating rate, the model uses Katayama's parameterization (1972). Predicted amount of water vapor and cloud are used in the computation as well as the prescribed amount of carbon dioxide. The dry convective adjustment, middle level convection and penetrative cumulus parameterization by Arakawa and Schubert (1974) are used for parameterizing convective processes. Surface fluxes of sensible heat, water vapor and momentum are computed by the bulk method with the transfer coefficient proposed by Deardorff (1972). Ground temperature, snow depth and ground wetness are predicted by considering ground thermodynamics and hydrology after Katayama. For further details of the model, see Tokioka et al. (1984).

Experiment

In the present experiment, the sun is fixed to the mean January position. Thus the model is under the perpetual January condition although the model atmosphere undergoes a diurnal cycle. Four cases of experiment are performed. We designate them as M, NM, NAS and NRG. In M, we use a realistic distribution of the earth's orography as shown in Fig. 1. In NM, we remove the whole mountains and set the height of land to the sea level. In NAS, we remove only the mountains over the Asian continent as hatched in 1, while, in NRG, we remove the mountains over North America and Greenland as stippled in 1.

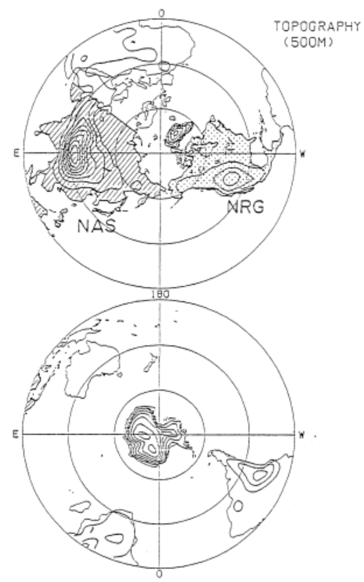

Figure 1: The distribution of topography adopted for the numerical experiments (M, NM, NAS and NRG). The contour interval is 500m. The orography removed in NAS and in NRG is hatched and stippled, respectively

annual march run at 00z of January 1. Those of the other runs are the same as that of M except that 1) the surface pressure ps and the surface temperature TS are extrapolated from those of M, 2) the ground temperature Tg is set equal to Ts, and 3) horizontal velocity, temperature and humidity fields are extrapolated and/or interpolated so as to take the same values at corresponding pressure heights of M. Time integration is made for 210 days in M. Data are sampled in every 8 hours. In other cases, time integrations are made for 200 days; the data in the first 50 days are discarded and the remaining 150 days' 8-hourly data are used for the following discussions.

Structure of stationary waves

Forcing fields

On comparing differences between the four model climates, we follow the argument by Held (1983). We shall refer to stationary eddy fields in NM as the "thermally forced" compomnent and the difference fields between the two stationary eddy fields as the "orographically forced" component by the difference between the respective orographies. Changing the pair , we can define five difference fields: (M-NM) , (M-NAS), (NRG-NM), (M-NRG) and (NASNM). Here, we regard the stationary eddy fields of M-NAS as the orographically forced component by the Asian mountains (especially, by the Tibetan Plateau). Similarly we regard those of NRG-NM as orographically forced components by mountains except the Rockies plus Greenland mountains; those of NAS-NM by the Rockies plus Greenland mountains; those of NAS-NM by mountains except the Asian mountains; and those of M-NM by all mountains.

The above separation of fields into the orographically and the thermally forced components is justified if the heating fields resemble one another between the four models (Held, 1983). 2 shows the vertically integrated diabatic heating fields and difference fields. Although differences are not small, the patterns are dominated by small-scale features in. M-NM and MNAS. As the geopotential field responses in less degree to smaller-scale heatings (Dickinson, 1980), the large-scale geopotential differences.

in those cases may not be much influenced by the small-scale heating differences. In M-NRG, however, heating differences are of large-scales and of the same order with the heating in NRG over the North Pacific and the North Atlantic

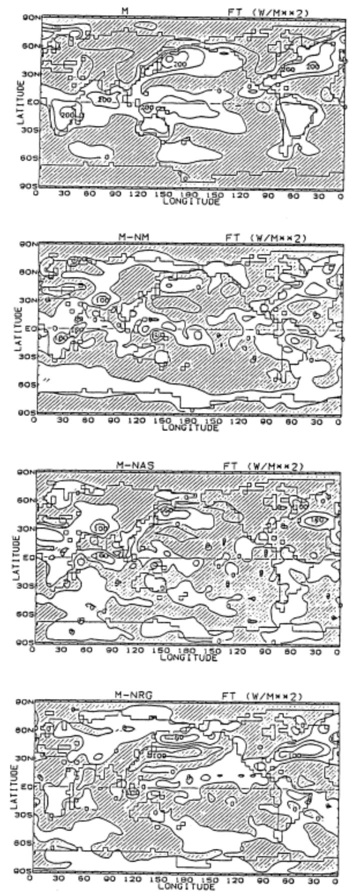

Figure 2: The vertically-integrated time-mean diabatic heating for M and the difference for M-NM , MNAS and M-NRG. The contour interval is 100 Wm-2 for M and 50Wm-2 for the rest. Negative values are hatched.

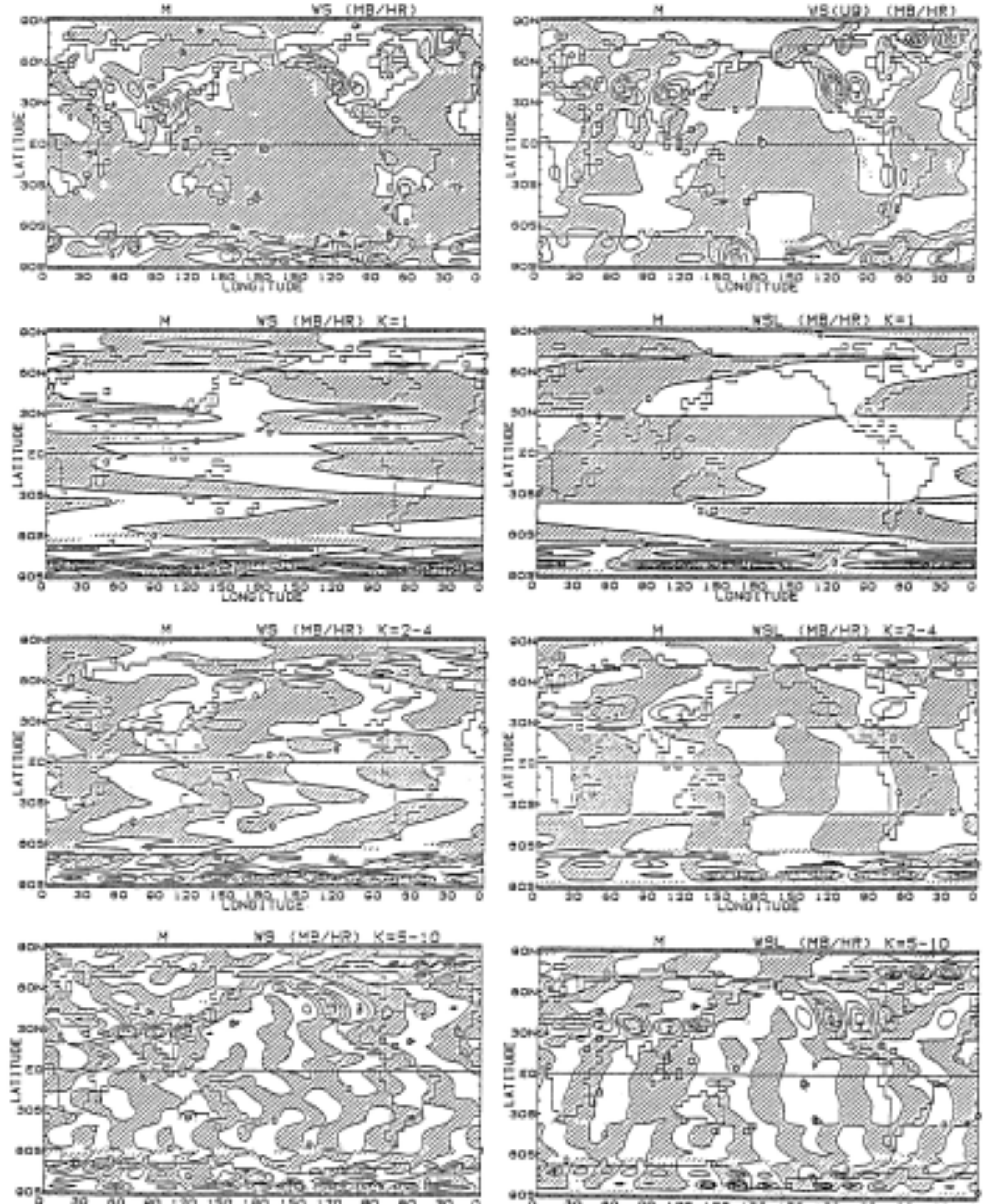

Figure 3: The surface p-velocity field *, (left) and *Ls given by Eq. (1) (right) for M. The original fields (top) are Fourier-decomposed into wavenumber 1 (second) , wavenumbers 2-4 (third) and wavenumbers 5-10 (bottom) components.

changes of surface flows caused by the dynamical effects of orography as will be discussed in the next subsection. In the following, however, we separate fields into the orographically and the thermally forced components rather in the loose sense.

In the linear models which are used for the investigation of atmospheric responses to orographic forcing, the perturbation vertical velocity induced by the flow over surface topography is usually given by:

$$\omega_i^L = \overline{U}\frac{\partial}{\partial x}p_s \qquad (1)$$

in pressure coordinates, where US is the zonally averaged tonal flow at the surface and x is a coordinate in the zonal direction. Figure 3 compares the surface p-velocity *s in M with *Ls where the zonally averaged 900mb zonal flow in M is adopted for Us. The top panels show the total fields, while the second, the third and bottom panels show the components of the zonal wavenumber 1, 2 to 4 and 5 to 10, respectively. Dipole patterns are not clearly seen around the Tibetan Plateau, Rockies and Greenland mountains in the total *s field. This reconfirms that diverted flows round mountains are not negligible (Saltzman and Irsch, 1972; Hayashi and Golder, 1977; Nakamura, 1978; Alpert et al., 1983; Egger, 1984). The position of the forced ascending (descending) motions *s are shifted northward (southward) from the positions expected in linear case. The intensity itself is not much reduced in *s from that in *Ls . As for the panel), the patterns are very different between and *Ls. The Eurasian Contient is *s covered by subsident motions in *s. On the other hand, the linear kinematical boundary condition requires ascending motion on the upwind side of the Tibetan Plateau. For the component of the wavenumbers 2 to 4, phases are compared well between *s and *Ls in the latitudes of 40*N to 60*N. However, non-negligible differences are found around 30*N, where subsidence is noted in *s at the southern slope of the Tibetan Plateau. This subsidence is associated with the anticyclonic flow around the Tibetan Plateau. The wavenumber components S to 10 mostly determine the characteristic distributions of ω_i . At this scale distributions of *s are not much different from the dipole patterns seen in ω_i.

Stationary waves in the Northern Hemisphere

5 shows the time mean sea-level pressure fields and the differences from that of M. The distribution of surface lows and highs are similar in location between the models, indicating that thermal effects are dominant near the surface. This feature is consistent with the previous results (Manabe and Terpstra, 1974; Held, 1983). However, orographic effects are not negligible. We see in M-NM a northward shift of the Siberian high and the high over North America due to orography. This change is also consistent with the results of Manabe and Terpstra (1974) and Nakamura (1978). A comparison between the N-NAS and MNRG fields shows that the Icelandic and Aleutian lows respond oppositely to the Tibetan Plateau forcing and to the Rockies plus Greenland forcing, i.e., the Tibetan Plateau (the Rockies plus Greenland mountains) strengthens (weaken) the Aleutian low but weakens (strengthen) the Icelandic low. In total, orography strenghens the Aleutian low as in Kikuchi (1979). However, the effects on the Icelandic low are not so clear. Corresponding to the sea level pressure differences in 4, there are also differences in the lower level winds, and thus in surface energy fluxes. The patterns found in the diabatic heating differences in 2 are qualitatively explained by the low level wind differences in the extratropics of the Northern Hemisphere, because the main heating there occurs in the regions where cold dry continental air flows over a warm ocean. In particular, we can see in M-NAS in 2 that the cold surge around the Tibetan Plateau strongly affects the heating over the Sea of Japan, the East China Sea and all the way from the Bay of Bengal to the Arabian sea. However, this effect found in the present experiment may be overestimated because of too strong anticyclonic flows around the Tibetan Plateau in M at the low level (see Tokioka et al., 1985). Figure 5 shows the temperature at 900mb. Increase of temperature is noted in polar regions in M. Uniform decrease of temperature occurs in mid to low latitudes. The temperature changes.

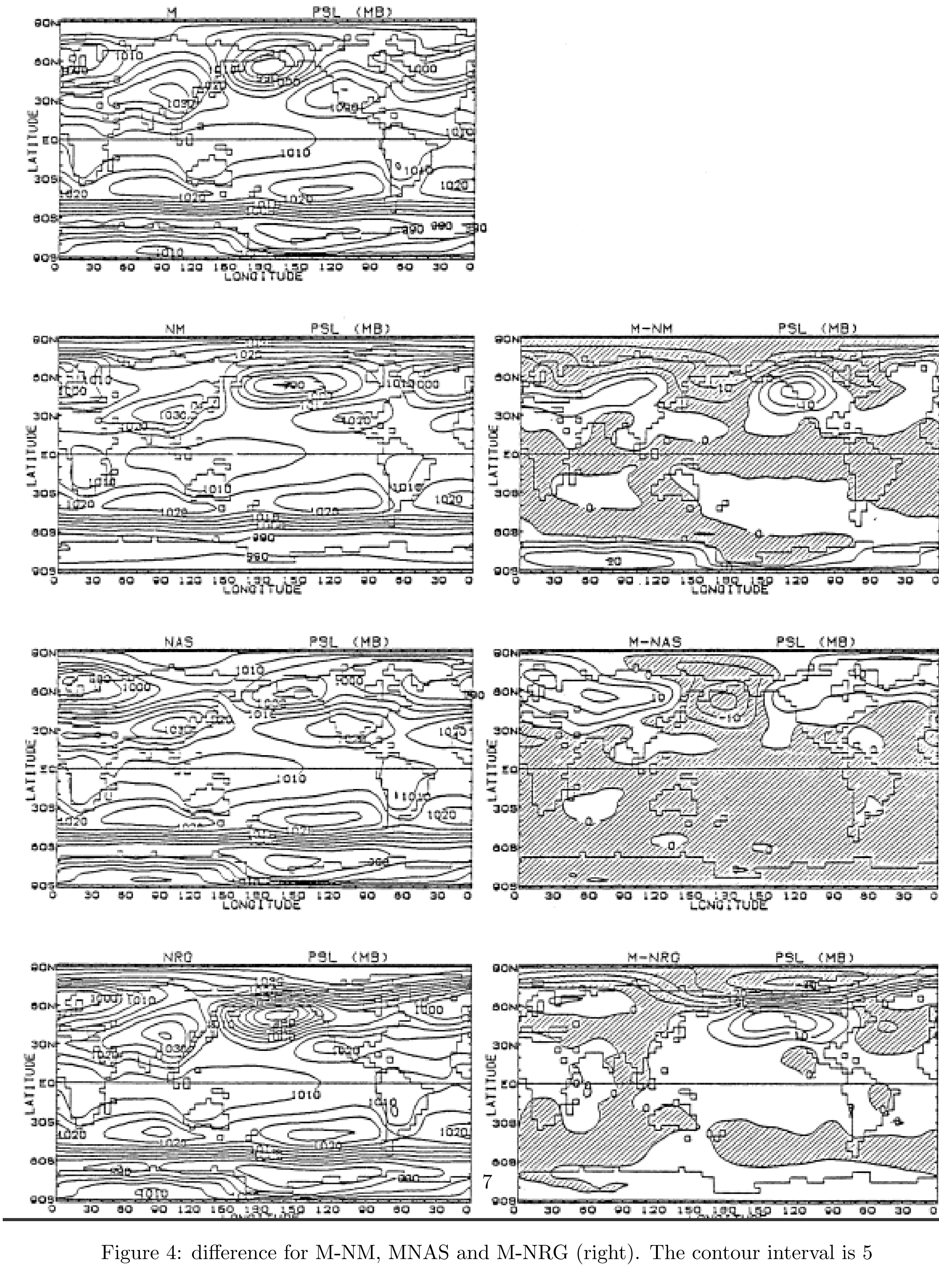

Figure 4: difference for M-NM, MNAS and M-NRG (right). The contour interval is 5

are in good correspondence with the changes of the low level wind. Along the eastern periphery of China, including Japan, temperature is decreased by the Tibetan Plateau due to pronounced cold winds. Nakamura and T. Murakami (1983) and T. Murakami and Nakamura (1983) have studied the mechanisms of cold surges in detail in relation to the lee-cyclogenesis.

A large temperature decrease is found in M-NAS over north Africa. The decrease of temperatures was initiated around day 45, when an intensified surface low passed there, bringing cold air mass from the Europian continent and also some amount of snow. Ever after, clouds were maintained in the lowest layer through the water vapor supply from the wet ground. Clouds prevented the ground from receiving shortwave radiation and kept the ground surface cold and wet. In addition to this hydrological effect, synoptic disturbances brought cold air mass more frequently compared with the other cases, aided with an eastward shift of upper trough over northwest Africa*. Figure 6 shows the 300mb time mean eddy geopotential height in M, NM and M-NM. As fround by Held (1983) and Ashe (1979), the thermal component (NM) and the orographic component (M-NM) are of the same order of magnitude. In NM, the zonal wavenumber 1 stationary wave is dominant in the Northern Hemisphere, although the small-scale structures are not negligible in the heating field in 2. This is because larger scale waves are enhanced in the thermal response (Dickinson, 1980). The wavenumber 1 component is extracted in Fig. 7. So far as this component is concerned, the maximum amplitude of the thermal component (NM) around 60*N is about three times as large as that of the orographic component (M-NM). As for the origin of the stationary wavenumber 1 in the Northern Hemisphere winter, controversy still exists. The conclusion that orographic component is most important above the middle

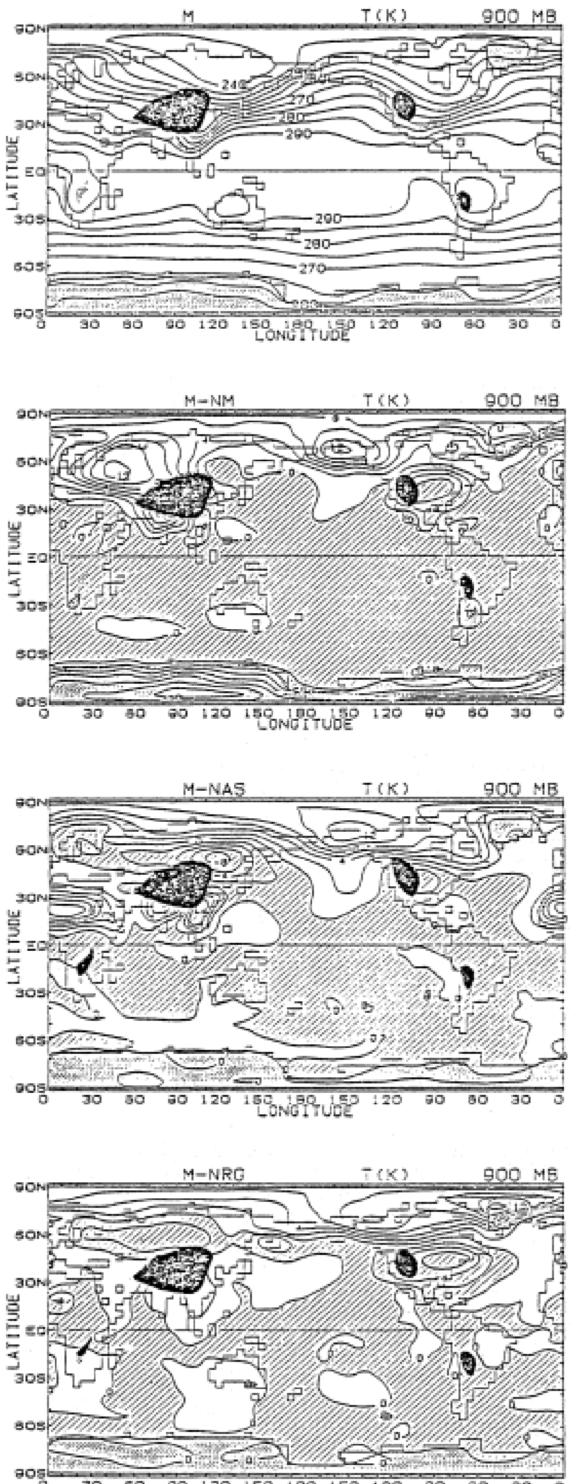

Figure 5: he time-mean temperature at 900mb for M (top) and the difference for M-NM (second), MNAS (third) and M-NRG (bottom). Units: K. Negative values are hatched.